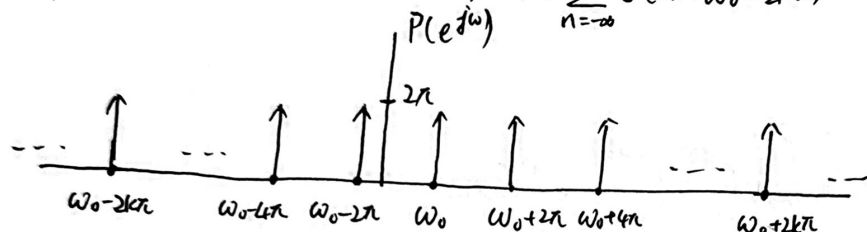


Problem 5.42

$$(a) \quad p[n] = e^{j\omega_0 n} \leftrightarrow P(e^{j\omega}) = 2\pi \sum_{n=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi n)$$



$$\begin{aligned} (b) \quad G(e^{j\omega}) &= \frac{1}{2\pi} \int_{<2\pi>} X(e^{j\theta}) \cdot 2\pi \sum_{n=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi n) d\theta \\ &= \int_{\omega - \omega_0 - \pi}^{\omega - \omega_0 + \pi} \sum_{n=-\infty}^{+\infty} X(e^{j(\omega - \omega_0 - 2\pi n)}) \delta(\omega - \omega_0 - 2\pi n - \theta) d\theta \\ &= X(e^{j(\omega - \omega_0)}) \int_{\omega - \omega_0 - \pi}^{\omega - \omega_0 + \pi} \sum_{n=-\infty}^{+\infty} \delta(\omega - \omega_0 - 2\pi n - \theta) d\theta \\ &= X(e^{j(\omega - \omega_0)}) \end{aligned}$$

Problem 5.52

$$(a) \quad h[n] = \begin{cases} 0, & n < 0 \\ h[0], & n = 0 \\ 2\text{Ev}\{h[n]\}, & n > 0 \end{cases} = 2\text{Ev}\{h[n]\} u[n] - \delta[n] \text{Ev}\{h[0]\}$$

$$\begin{aligned} H(e^{j\omega}) &= \frac{1}{\pi} \int_{<2\pi>} \text{Re}\{H(e^{j\theta})\} \left(\frac{1}{1 - e^{j(\omega - \theta)}} + \pi \sum_{n=-\infty}^{+\infty} \delta(\omega - \theta - 2\pi n) \right) d\theta - \text{Ev}\{h[0]\} \sum_{n=-\infty}^{+\infty} \delta[n] e^{-j\omega n} \\ &= \frac{1}{\pi} \int_{<2\pi>} \frac{\text{Re}\{H(e^{j\theta})\}}{1 - e^{j(\omega - \theta)}} d\theta + \text{Re}\{H(e^{j\omega})\} - \frac{1}{2\pi} \int_{<2\pi>} \text{Re}\{H(e^{j\omega})\} d\omega \end{aligned}$$

$$(b) \quad \text{Re}\{H(e^{j\omega})\} = 1 + \alpha \cos 2\omega$$

$$\begin{aligned} \text{Ev}\{h[n]\} &= \frac{1}{2\pi} \int_{<2\pi>} (1 + \alpha \cos 2\omega) e^{j\omega n} d\omega = \begin{cases} 1, & n = 0 \\ \frac{\alpha}{4\pi} \int_{<2\pi>} (e^{j(n-2)\omega} + e^{j(n+2)\omega}) d\omega, & n \neq 0 \end{cases} \\ &= \begin{cases} 1, & n = 0 \\ \frac{\alpha}{2}, & n = \pm 2 \\ 0, & \text{otherwise} \end{cases} \quad h[n] = \begin{cases} 0, & n < 0 \\ 1, & n = 0 \\ \alpha, & n = 2 \\ 0, & \text{otherwise} \end{cases} = \delta[n] + \alpha \delta[n-2] \\ H(e^{j\omega}) &= 1 + \alpha e^{-j(2\omega)} \end{aligned}$$

$$(c) \quad h[n] = \begin{cases} 0, & n < 0 \\ h[0], & n = 0 \\ 2\alpha\{h[n]\}, & n > 0 \end{cases} = 2\alpha\{h[n]\} u[n] + \delta[n] h[0]$$

$$H(e^{j\omega}) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{1}{j\omega} j \text{Im}\{H(e^{j\theta})\} \left(\frac{1}{1 - e^{j(\omega-\theta)}} + \pi \sum_{n=-\infty}^{\infty} \delta(\omega-\theta-2\pi n) \right) d\theta + h[0]$$

$$= \frac{j}{\pi} \int_{-\infty}^{\infty} \text{Im}\{H(e^{j\theta})\} \cdot \frac{1}{1 - e^{j(\omega-\theta)}} d\theta + j \text{Im}\{H(e^{j\omega})\} + h[0]$$

$$(d) \quad \text{Im}\{H(e^{j\omega})\} = \sin \omega$$

$$\alpha\{h[n]\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} j \sin \omega e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{j\omega} - e^{-j\omega}}{2j} \cdot e^{j\omega n} d\omega$$

$$= \frac{1}{4\pi} \int_{-\infty}^{\infty} (e^{j(n+1)\omega} - e^{j(n-1)\omega}) d\omega = \begin{cases} \frac{1}{2}, & n = -1 \\ -\frac{1}{2}, & n = 1 \\ 0, & \text{otherwise} \end{cases} = \frac{1}{2}(\delta[n+1] - \delta[n-1])$$

$$h[n] = (\delta[n+1] - \delta[n-1]) u[n] + \delta[n] h[0] \quad (\text{解})$$

$$h_1[n] = (\delta[n+1] - \delta[n-1]) u[n], \quad h_2[n] = (\delta[n+1] - \delta[n-1]) u[n] + \delta[n]$$

5.53

$$(a) \quad \tilde{X}[k] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j k \omega_0 n} = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j (\frac{2\pi k}{N}) n} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j (\frac{2\pi k}{N}) n}$$

$$= \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-j (\frac{2\pi k}{N}) n} = \frac{1}{N} X(e^{j(\frac{2\pi k}{N})})$$

$$(b) \quad X_1(e^{j(\frac{2\pi k}{4})}) = \sum_{n=-\infty}^{\infty} x[4n] e^{-j (\frac{2\pi k}{4}) n} + \sum_{n=-\infty}^{\infty} x[4n+1] \cdot (-j) e^{-j (\frac{2\pi k}{4}) n} = \sum_{n=0}^3 x[n] e^{-j (\frac{2\pi k}{4}) n} = 1 + (-1) \cdot \left(\frac{1}{j}\right)^k + 2 \cdot j^k$$

$$X_2(e^{j(\frac{2\pi k}{4})}) = (x_2[-2] + x_2[2] + x_2[6]) e^{-j(\frac{2\pi k}{4}) \cdot 2} + (x_2[-1] + x_2[3] + x_2[7]) e^{-j(\frac{2\pi k}{4})}$$

$$+ (x_2[0] + x_2[4]) \cdot 1 + (x_2[1] + x_2[5]) e^{-j(\frac{2\pi k}{4})}$$

$$= 0 + 2j^k + 1 + (-1) \cdot \left(\frac{1}{j}\right)^k = X_1(e^{j(\frac{2\pi k}{4})})$$